- Skid hazards
- Surface treatment distresses

Information from several sources was used to compile the information included in this section of the chapter.

Typical Causes of Deterioration

Some of the primary causes of HMA pavement deterioration are listed here:

- Traffic loading
- Environment or climate
- Drainage deficiencies
- Material problems
- Construction deficiencies
- External causes (such as utility cuts)

As pavements age, one or more of these distress mechanisms begins to take its toll. As cracking and other forms of disintegration begin to appear as the primary cause of deterioration, secondary factors often contribute additional amounts of deterioration. For instance, once cracking begins to appear in the pavement surface, moisture can enter the pavement structure and accelerate the deterioration caused by the initial distress mechanism. The timely application of pavement maintenance techniques serves to help prevent or slow down the effects of both primary and secondary distress mechanisms.

Cracking

A number of different types of cracking can occur in an HMA pavement and the appropriate type of repair depends on the type of cracking present. Some cracking, such as alligator cracking, indicates load-related deterioration that requires a different maintenance strategy than block cracking, which is typically caused by climatic forces. This section provides information on some of the most common types of cracks included in a pavement evaluation.

Alligator (Fatigue) Cracking

Alligator, or fatigue, cracking appears on a pavement surface as a series of interconnected cracks that resemble the skin of an alligator. Alligator cracking indicates load-related deterioration because it is caused by fatigue failure of the HMA surface under repeated traffic loadings and is often related to weakening of the base course or subgrade, insufficient pavement thickness, excessive loading, or some combination of these factors. An example of alligator cracking is shown in Figure 11.9.

Traditionally, alligator cracking starts at the bottom of the asphalt layer, where the tensile stress and strain under a wheel load is highest. The cracks propagate up to the pavement surface where they first appear as parallel longitudinal cracks in the wheelpaths. Under repeated traffic loadings, further deterioration takes place and the cracks begin to interconnect, forming many angular pieces whose appearance is similar to that of an alligator's skin or chicken wire.

In recent years, it has begun to be recognized that top-down fatigue cracking may occur in HMA pavement. These commonly occur in thicker pavements that are subjected to large horizontal tensile stresses developing in the surface

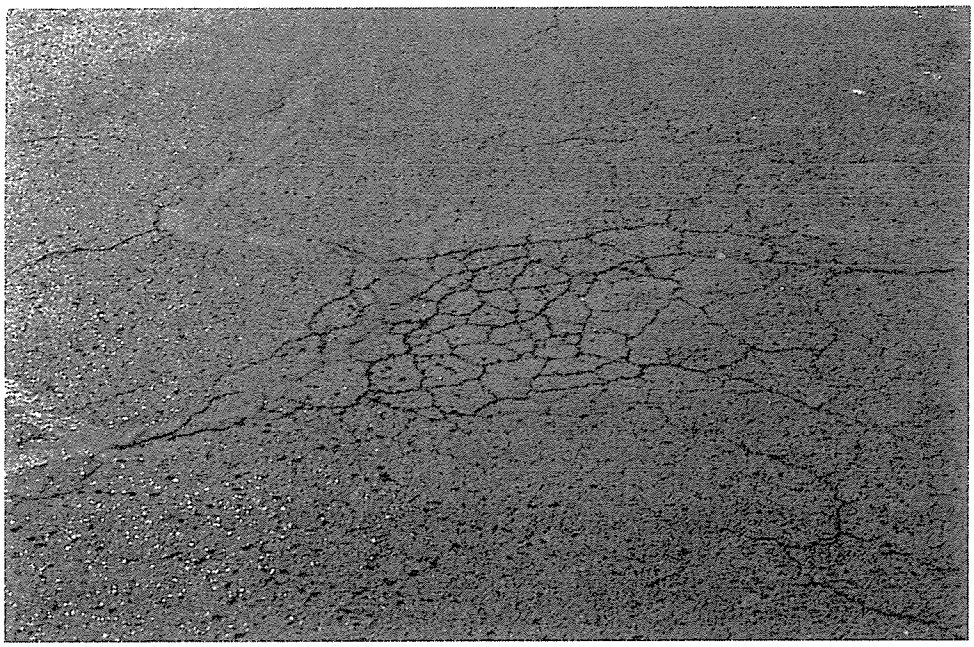


Figure 11.9 Alligator (Fatigue) Cracking

due to high tire contact pressures. Therefore, to determine whether a crack is a top-down crack or a bottom-up crack, it may be necessary to core the pavement surface before selecting a maintenance strategy.

Block Cracking

Block cracks are interconnected cracks that form a series of large, rectangular pieces on the pavement surface that are typically greater than one foot (0.3 meter) in each direction. Block cracking is usually found when the HMA surface has hardened and the daily temperature cycles cause shrinkage in the HMA layer; however, it may also be caused by volume changes in a fine-aggregate HMA mix containing low-penetration asphalt and adsorptive aggregates. Low traffic levels can aggravate this latter situation because the restorative working properties of asphalt require frequent traffic applications.

Block cracking typically covers a large area of pavement, including areas without traffic. The size of the blocks and the presence of block cracking in nontrafficked areas help to differentiate block cracking from alligator cracking in the field. An example is provided in Figure 11.10.

Edge Cracking

Longitudinal cracks within one to two feet of the outer edge of a pavement are referred to as edge cracks. Over time, as vehicles pass over the weakened area, further break-up of longitudinal cracks may take place, resulting in crescent-shaped cracks that branch to the edge of the pavement. Edge cracking is illustrated in Figure 11.11.

Edge cracks form along a weakened pavement edge that does not have adequate lateral support from a shoulder. The weakened pavement edge



Figure 11.10 Block Cracking



Figure 11.11 Edge Cracking

can be caused by a number of factors, including a weak base or subgrade layer, poor drainage, or frost heaves. Heavy traffic along the pavement edge or heavy vegetation along the edge of the pavement can accelerate the progression of edge cracking.

Longitudinal (Linear) and Transverse Cracking

This category of distress includes linear cracks in the pavement surface that are either parallel or perpendicular to the centerline of the road, with the exception of linear cracks that are in the wheelpaths (which are considered fatigue cracks). Longitudinal cracks are parallel to the pavement centerline and may be caused by the following:

- A poorly constructed paving joint crack, as shown in Figure 11.12
- Shrinkage of the asphalt layer
- Daily temperature cycling
- Cracks in an underlying layer that reflect up through the pavement (with the exception of joint reflection cracking, which is distinguished as a separate type of crack)
- Longitudinal segregation caused by the improper operation of the paver

Transverse cracks, which are perpendicular to the centerline, can also be caused by shrinkage in the asphalt layers or by cracking in an underlying layer. An example of transverse cracking in shown in Figure 11.13. Longitudinal and transverse cracks are not considered load-related cracks.

Reflection Cracking (PCC Joint Reflection Crack)

Reflection cracking includes cracks that propagate through the HMA layer of an overlaid Portland cement concrete (PCC) pavement at its joints, or cracks that propagate through an overlay over a deteriorated asphalt surface.

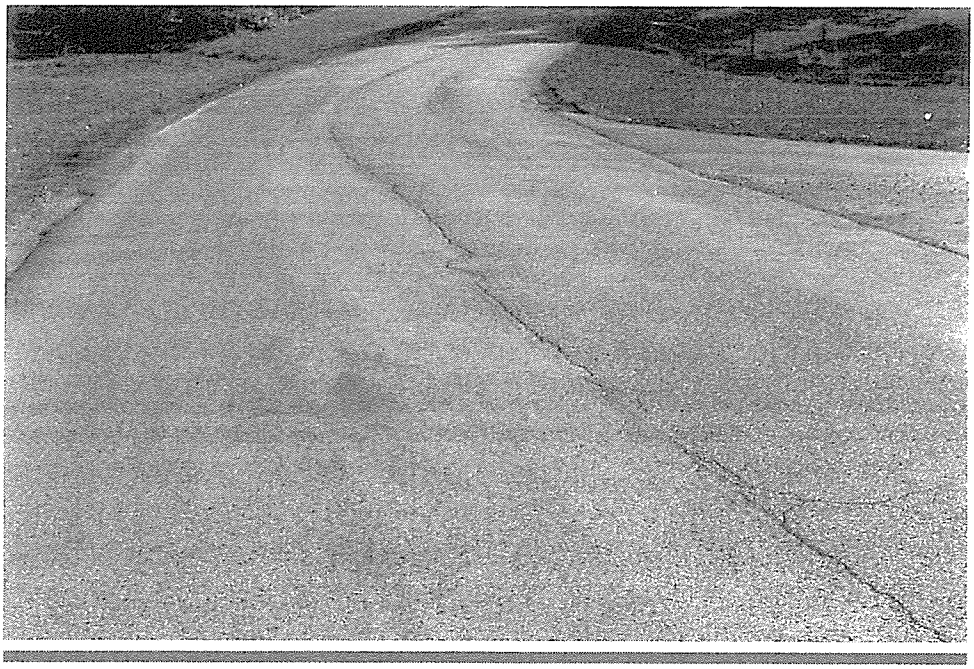


Figure 11.12 Longitudinal Cracking

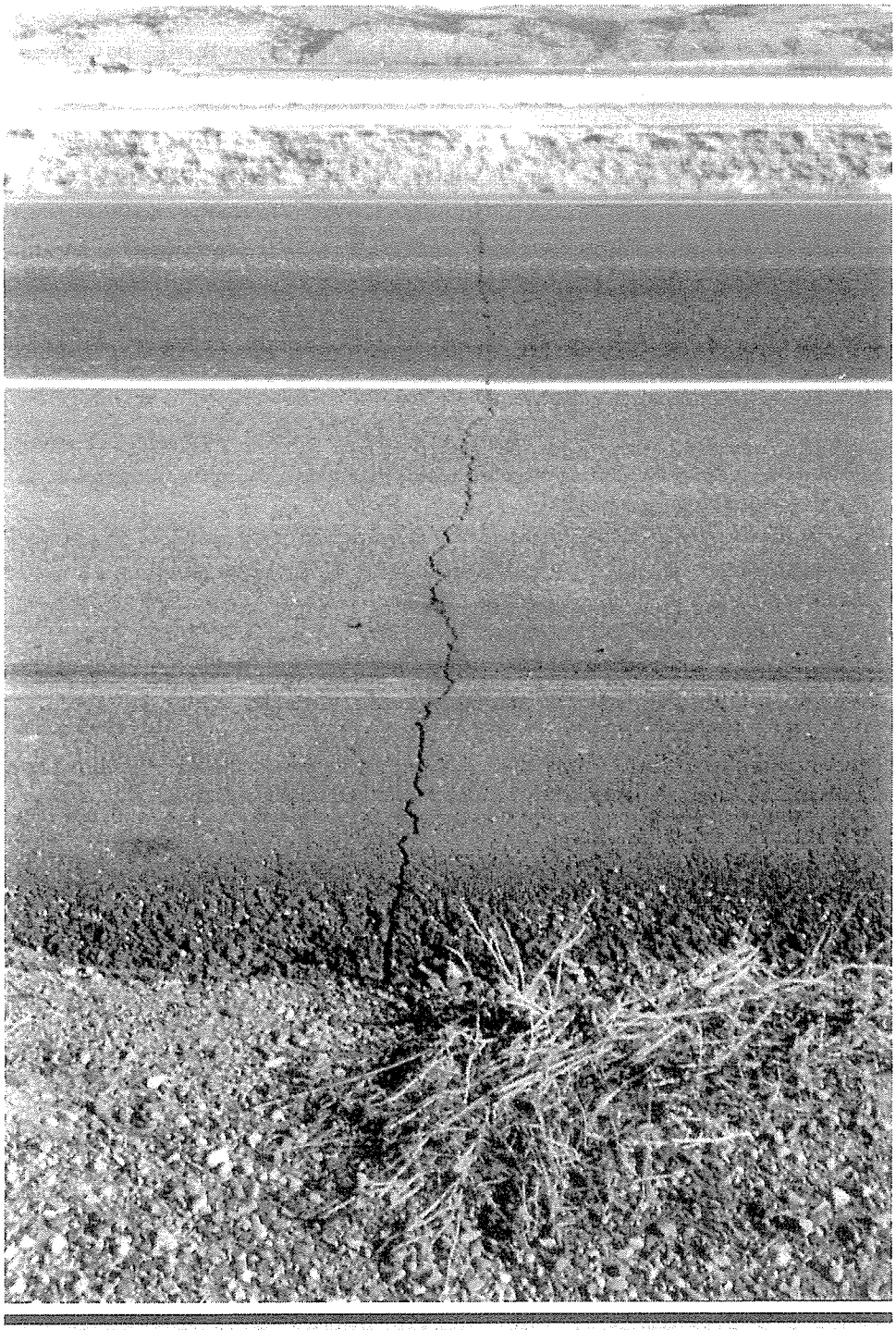


Figure 11.13 Transverse Cracking

Reflection cracks are initially caused by differential movement between the asphalt and concrete layers, but the cracks can deteriorate further under heavy traffic. It is helpful to know the slab dimensions of the PCC pavement to identify this distress. This distress is illustrated in Figure 11.14.

Slippage Cracking

Slippage cracks are easily identified by their distinctive crescent- or half-moon-shape cracks, as illustrated in Figure 11.15. They are usually caused by vehicular turning or stopping movements in pavements with a low-strength surface mix or a poor bond between pavement layers. The cracks are caused by the horizontal forces of stopping or turning vehicles, so the ends of the curved crack point in the opposite direction of the vehicle thrust.

Distortion

Distortions in the pavement layer are a result of instability in the HMA layer or weaknesses in the base or subgrade layers. The distortion may appear in a number of different forms, including rutting, shoving, corrugations, depressions, or upheavals. Cracking may or may not accompany the distortion. Each of the typical types and causes of distortions are discussed further in this section.

Corrugations and Shoving

Corrugations and shoving, also known as washboarding, result in ripples across the asphalt pavement surface at fairly regular intervals of less than ten feet (3 meters) and perpendicular to the traffic direction. Corrugations and shoving typically occur at points of severe horizontal stress

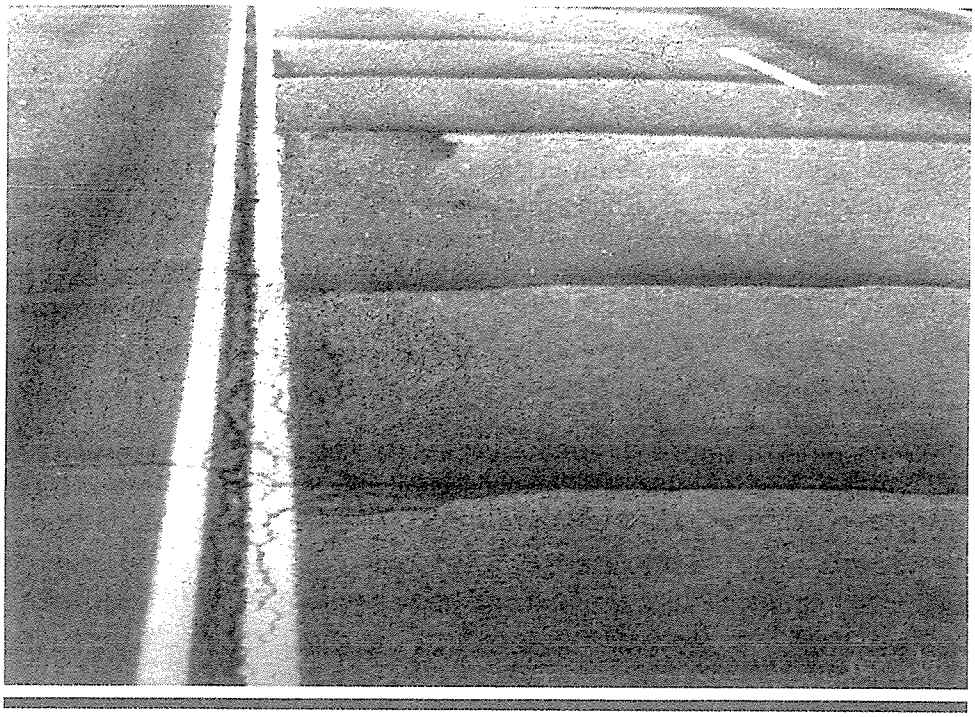


Figure 11.14 Reflection Cracking



Figure 11.15 Slippage Cracks

(such as locations where traffic starts and stops) in HMA layers that lack mix stability. Distortions can be caused by a number of factors, including mixtures that are too rich in asphalt or too high in fine-aggregate content, aggregates that are too smoothly textured, asphalt cement that is too soft, or a granular base with excessive moisture or contamination. An example of shoving is shown in Figure 11.16.

Rutting

A rut is a surface depression in the wheelpaths that may also have transverse displacement along the sides of the rut. Rutting is caused by consolidation or lateral movement of any of the pavement layers or the subgrade under traffic. It may be caused by insufficient design thickness, lack of compaction, weaknesses in the pavement layers due to moisture infiltration, or weak asphalt mixtures. Rutting is illustrated in Figure 11.17.

Settlement or Grade Depressions

Depressions and settlements are localized pavement surface areas with elevations lower than the surrounding pavement area. Minor depressions are often not noticed until after a rain when water ponds in the depression, causing a "birdbath" that can be hazardous to motorists. In dry weather, depressions can be observed where staining is present. Depressions may be caused by settlement or failure in the lower pavement layers or by improper construction techniques. An example of a depression is shown in Figure 11.18.

Upheaval or Swell

An upheaval or swell is a localized upward displacement in a pavement due to swelling of the subgrade or some portion of the pavement structure.



Figure 11.16 Shoving

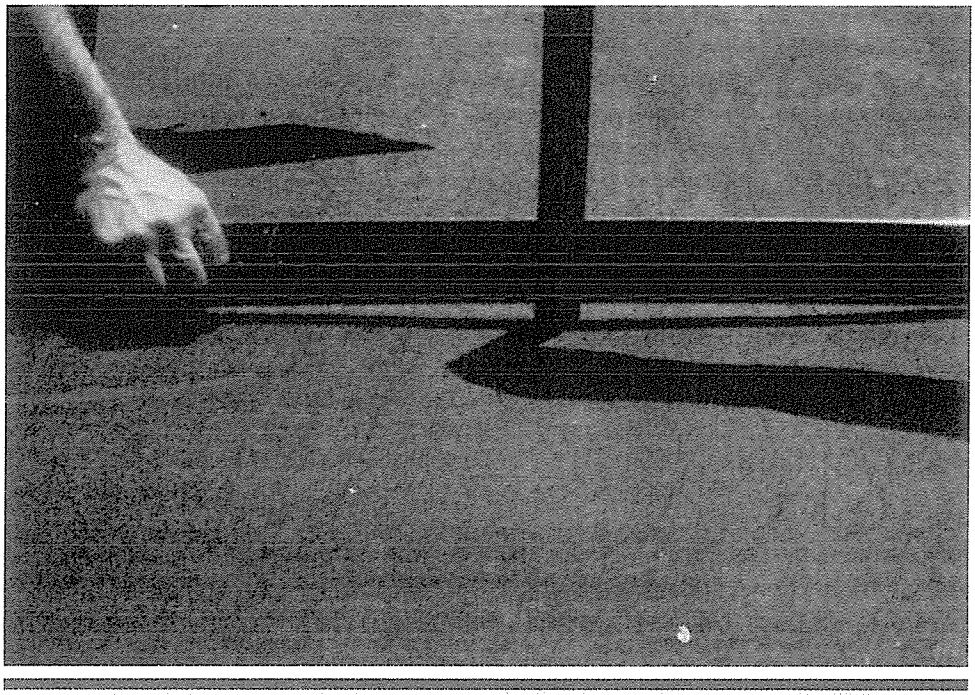


Figure 11.17 Rutting

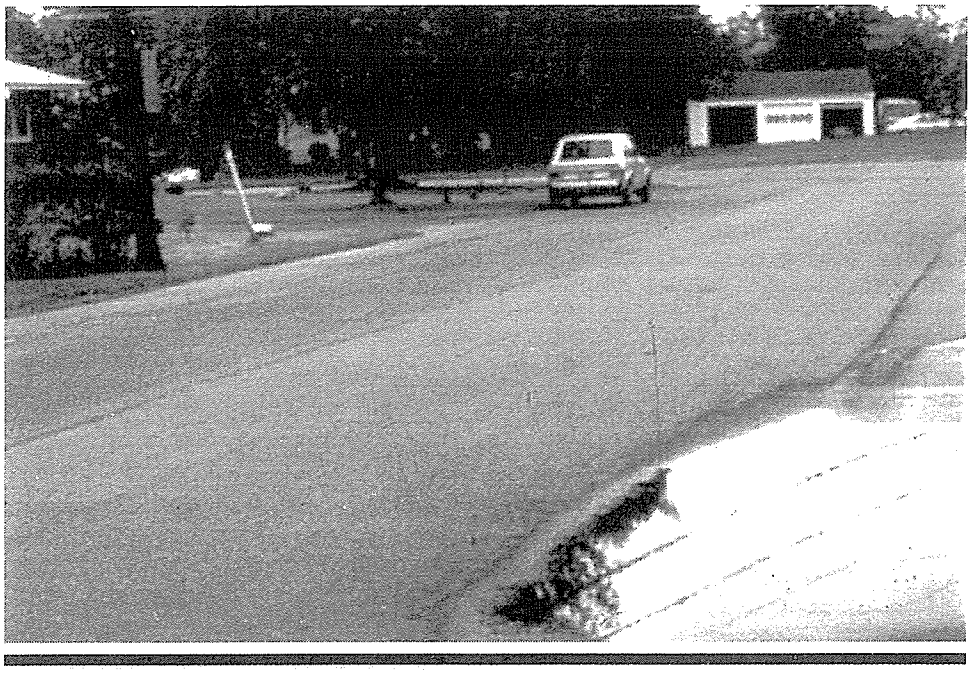


Figure 11.18 Depression

A frost heave is an example of this type of distress, as is the upheaval shown in Figure 11.19.

Upheavals, or swells, are often the result of expansive soils (which swell in the presence of moisture) or frost heave (in which ice lenses grow beneath the pavement, causing the pavement to crack).

Utility Cuts and/or Patch Failure

A patch is a portion of a pavement that has been removed and replaced or where additional material has been added. Patches are a form of pavement distress regardless of how well they are performing, but the severity of the problem increases when the patch has deteriorated. The rate at which a patch deteriorates may be influenced by poor installation techniques, such as inadequate compaction, inferior materials, or failure of the surrounding or underlying pavement. An example of a utility patch is provided in Figure 11.20.

Disintegration

Disintegration includes the breakup of the pavement into smaller pieces that may become dislodged over time. It is important to address pavement disintegration early, before too much of the pavement surface has been lost. Weathering, raveling and potholes are examples of pavement disintegration.

Potholes

A pothole is a bowl-shaped hole caused by localized disintegration of the pavement surface, as shown in Figure 11.21. Potholes are typically caused by the continued deterioration of another type of distress, such as alligator



Figure 11.19 Upheaval

cracking, raveling, poor drainage, or a failed patch after pieces of the original pavement surface have been dislodged. Poor surface mixtures, weak spots in the base or subgrade, or the severity of the surrounding distress accelerate potholes. Potholes are considered a structural form of deterioration.

Raveling and Weathering

As pavements ravel and weather, the pavement surface wears away because of a separation of the aggregate particles from the asphalt binder. The aggregate typically separates first, leaving *pockmarks* in the surface. As more particles dislodge, the pavement becomes rougher and further deterioration takes place until large pieces have been displaced. Raveling is usually found in the wheelpath, while weathering more commonly covers the entire pavement surface. A raveled surface is illustrated in Figure 11.22.

The presence of this distress indicates that either the asphalt binder has hardened appreciably or that a poor-quality mixture is present. Raveling usually requires the presence of both traffic and water to occur.

Skid Hazards

An important component of providing a safe pavement surface is keeping the surface free of properties that might increase the likelihood of vehicles skidding on the surface, such as surface water, polished aggregates, or excess oil on the pavement surface. To address these issues, it is important to restore the pavement surface in a way that allows water to flow off the pavement surface and maintains sufficient contact between the vehicle tires and aggregate.

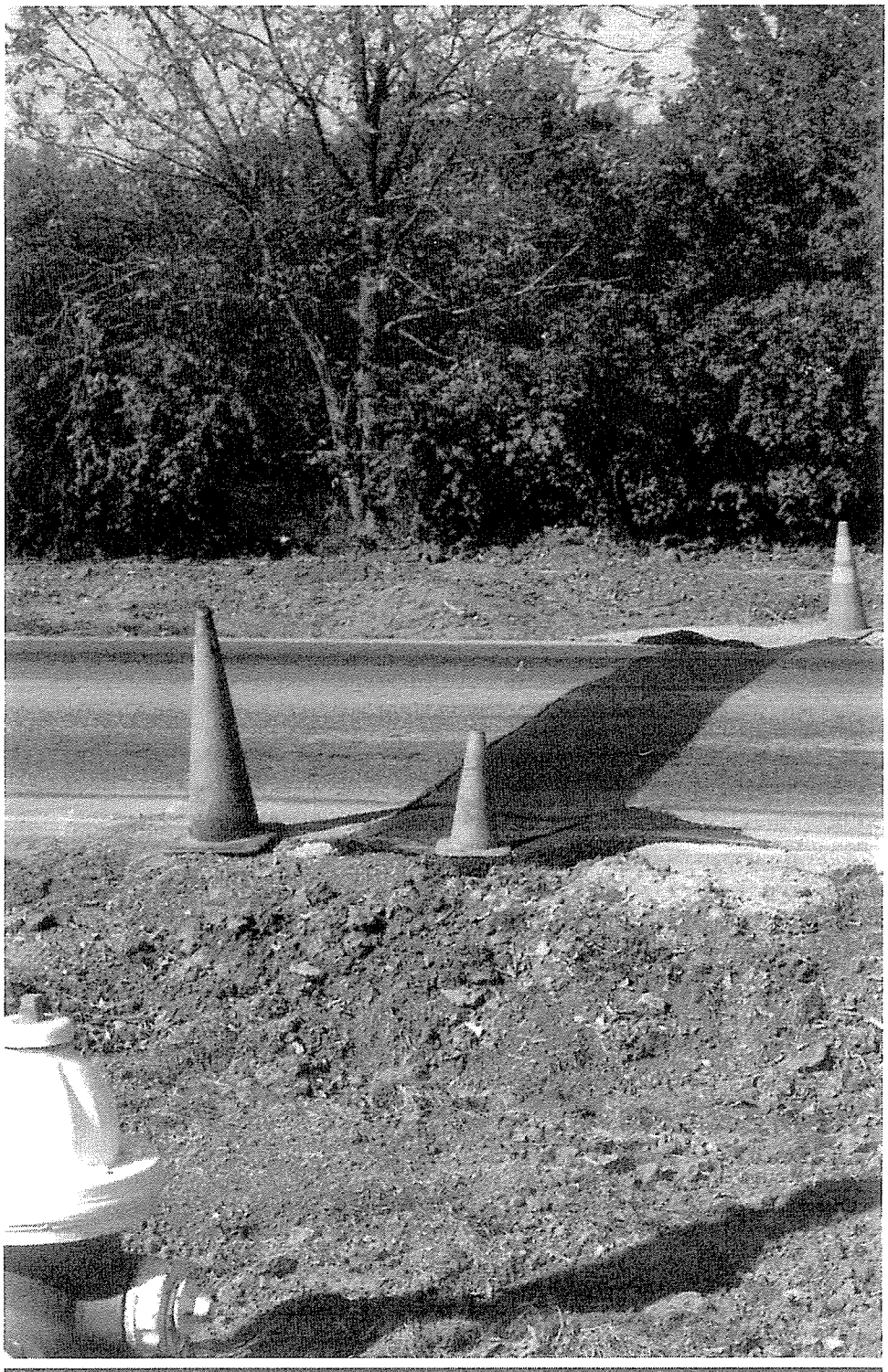


Figure 11.20 Utility Cut

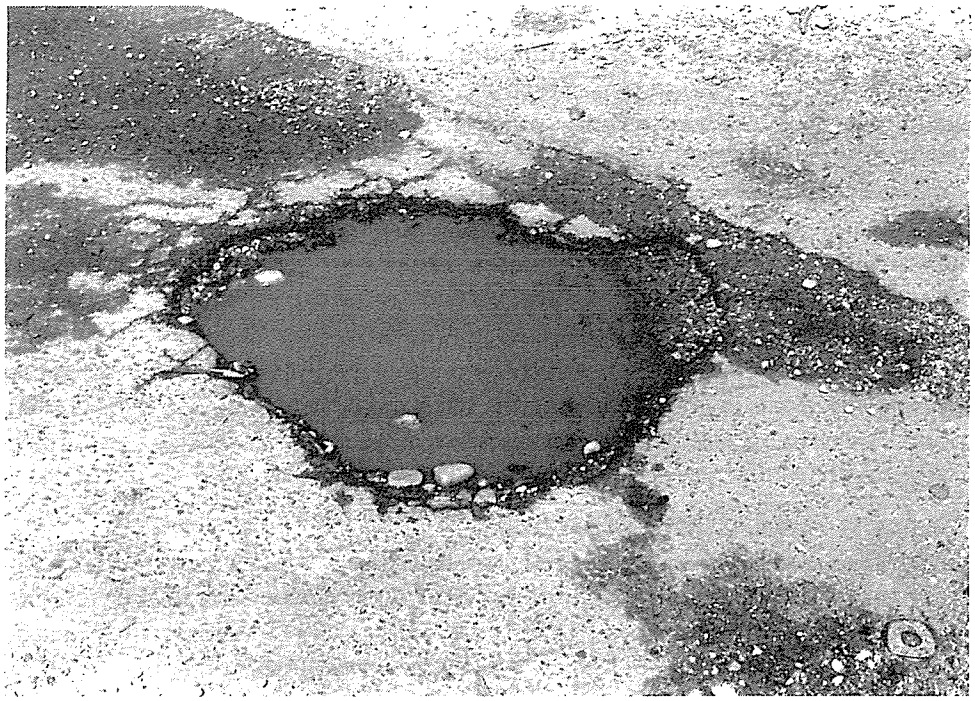


Figure 11.21 Potholes



Figure 11.22 Raveling

Bleeding or Flushing

Bleeding, or flushing, leaves a thin film of asphalt on the pavement surface resulting from excess bitumen. Bleeding is easy to identify in the field by its slick, glossy appearance and the tendency of the asphalt to often be sticky to the touch (especially in hot weather), as illustrated in Figure 11.23. Bleeding can be caused by an improperly constructed seal coat, too much asphalt in a mix, too heavy a prime or tack coat, or excessive sealant in the cracks or joints under an overlay. Traffic can also cause bleeding if the asphalt layers are overcompacted and the binder is forced to the surface.

Polished Aggregate

Aggregates that have been worn to a smooth finish are called polished aggregates. Some aggregates, such as limestone, may polish quickly under traffic. Once the aggregates polish, they are a potential skid hazard, especially when wet. An example of a polished surface is provided in Figure 11.24.

Surface Treatment Distress

Because of their unique properties, some types of distress may occur only in an asphalt surface treatment and not in other types of asphalt pavements, such as loss of cover aggregate and streaking. Other distresses discussed earlier in this chapter, such as depressions, raveling, corrugations, and potholes occur in both hot-mix asphalt and bituminous surface treatments.

Loss of Cover Aggregate

When the aggregate in a surface treatment is lost earlier than expected, a loss of cover aggregate occurs. This can be caused if the aggregate is spread

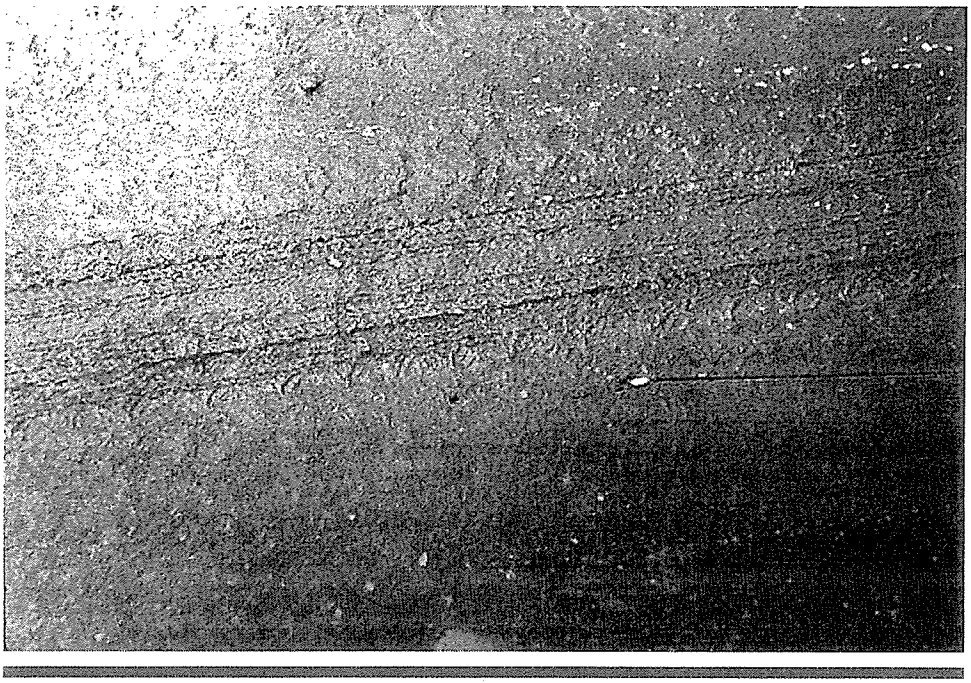


Figure 11.23 Bleeding



Figure 11.24 Polished Aggregate

too late after the binder is applied, if the aggregate is too dirty and does not bond with the asphalt emulsion, if traffic is applied too soon after the placement of the surface treatment, or if there is too little asphalt to hold the aggregate. Improper compaction with a steel-wheel roller alone may also cause a loss of aggregate in low spots that are bridged by the roller. An example of a pavement section with a loss of cover aggregate is provided in Figure 11.25.

Longitudinal/Transverse Streaking

Longitudinal streaking is observed in pavement sections with alternating lean and heavy lines of asphalt or aggregate that run parallel to the centerline of the road, as shown in Figure 11.26. A number of factors can cause longitudinal streaking, including the following:

- The spray bar on the asphalt distributor is not set at the proper height so spray fans do not overlap properly.
- The spray bar rises as the load in the distributor lightens.
- The nozzles on the spray bar are not at the same angle, are not the right size, or are plugged with asphalt.
- The asphalt is too cool.
- The pump pressure is incorrect.

Transverse streaking is similar in appearance to longitudinal streaking, except that the streaking runs transverse to the centerline. Transverse streaking is caused by the uneven distribution of asphalt spray from the distributor spray bar.



Figure 11.25 Loss of Cover Aggregate

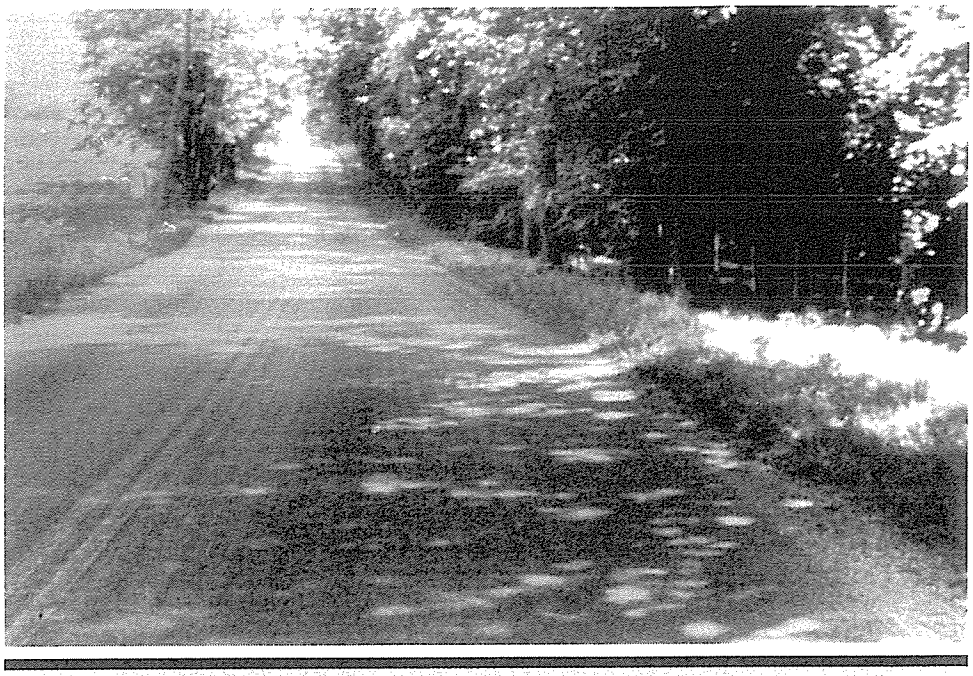


Figure 11.26 Longitudinal Streaking

11.4 Pavement Preservation Program

As discussed earlier in this chapter, a pavement preservation program can be a cost-effective way to manage a pavement network when a blend of preventive maintenance, rehabilitation, and reconstruction treatments is used appropriately. This section of the chapter describes the components of an effective pavement preservation program in more detail and the manner in which a pavement preservation program is supported using a pavement management system.

11.4.1 Establishing a Pavement Preservation Program

The decision to establish a pavement preservation program can be difficult for an agency because it often requires a change in the way an agency does business. Traditionally, agencies respond to pressure from outside forces to continually allocate funds to the roads in the worst condition. However, when a strong pavement preservation program is in place, in addition to funding roads in poor condition, the agency dedicates some of its funds to functional improvements on roads that are not substantially deteriorated in order to delay the need for more costly treatments. It requires support from the top of the organization to commit to this type of program and to remain committed in the face of outside pressures to do otherwise.

Use of Pavement Management to Support Pavement Preservation

The use of pavement management systems to demonstrate the benefits associated with a pavement preservation program was discussed earlier in this chapter. To fully support a comprehensive pavement preservation program, a pavement management system must include the following considerations in its analysis:

- A method of assessing pavement conditions that includes the types of deterioration that normally trigger the selection of preventive maintenance and rehabilitation treatments
- Pavement performance models that can differentiate the performance of a pavement section with and without different types of treatments
- Treatment rules that can be used to determine when different types of treatments are considered feasible
- Models that quantify the effects on pavement condition after different types of treatments have been applied. These models are used to forecast future conditions under various scenarios.

Developing a Program

A pavement preservation program can be initiated with a decision to allocate funds to preservation treatments at different levels depending on the pavement conditions. The appropriate amount of funds allocated to preventive maintenance, rehabilitation, and reconstruction activities depends on the current and projected condition of the agency's pavement network. However, using the pavement management system, an agency can evaluate different funding allocations to determine which strategy comes closest to meeting the agency's targeted conditions. Agencies that have been most successful with their pavement preservation program have initiated other activities to support the preservation program, including training at all levels within the organization, the establishment of dedicated funds for preventive maintenance treatments, and the establishment of guidelines that help ensure that treatments are consistently used effectively throughout the organization.

Estimating Needs

The pavement management system can be used to identify funding needs based on current and projected pavement conditions. To estimate needs, an agency must first establish either a minimum condition level (above which the network should be maintained) or a targeted condition level for each road classification (such as arterials, collectors, and local roads). Using the pavement condition information in the pavement management system, the number of miles of roads that do not meet these condition levels can be estimated along with the cost of repair at these levels. Together, this information is used to estimate the total pavement needs of the agency. By tracking the change in pavement needs over time, an agency can quickly determine if it is "catching up" or "falling behind."

Setting Goals

Information from the pavement management system can also be used to set performance targets for the pavement network. This is typically an iterative process in which anticipated funding levels are input into the pavement management system and expected condition levels at a particular point in time are calculated. By varying the funding levels input into the analysis, a realistic estimate of future conditions can be determined and a targeted goal can be established. In most cases, different pavement condition levels are established for different road classifications in a typical network, reflecting the differences in traffic levels, traffic speeds, and customer expectations. For example, a lower targeted condition might be established for local streets than for arterial roads that serve a larger number of vehicles traveling at higher traffic speeds.

The establishment of a goal is an important part of a pavement preservation program because it helps the agency measure the program's effectiveness. The best goals are based on information from the pavement management system, can easily be remembered, define the expected conditions, and set the timeframe for achieving the targets.

11.4.2 Selecting the Appropriate Treatment

One of the primary uses of a pavement management system is the identification of appropriate maintenance and rehabilitation treatments to address the types and the extent of pavement deterioration. A pavement management system can recommend either the appropriate level of repair or a treatment. Each approach is described in this section.

Determining the Appropriate Level of Repair

In some agencies the pavement management system is used to determine the level of repair needed rather than the specific treatment that might be recommended. For instance, a pavement section might be recommended as a candidate for preventive maintenance, minor rehabilitation, major rehabilitation, or reconstruction, depending on the results of the pavement condition survey and the agency's treatment policies. Average cost information is associated with each level of repair so the estimated cost of each project can be determined. Average rates of deterioration for each treatment are used to predict future conditions. While this approach is fairly easy to set up and use in the pavement management system, the use of average cost and performance data may not provide the level of accuracy desired by some agencies. Agencies that prefer to conduct more detailed project-level pavement evaluations before selecting the appropriate treatment might prefer to use this approach in their network-level pavement management analysis.

Selecting Appropriate Treatments

An alternate approach is to define specific treatments in the pavement management system so the analysis can determine the optimal treatment and timing for each project. This approach requires more detail in setting up the pavement management system because treatment rules, performance models, and impact rules must be established for each potential treatment. Treatment rules define the feasibility of a treatment, performance models approximate the rate of deterioration after a treatment has been applied, and impact rules define the changes that take place after a treatment has been applied (for example, any change in surface type or a change in pavement condition).

Two approaches used to select appropriate maintenance and rehabilitation trees—decision trees and decision matrices—are discussed here in more detail.

Decision Trees

Some pavement management systems use decision trees to determine feasible treatment options. As the name implies, options are differentiated by different branches to the tree. For example, a very simple decision tree is presented in Figure 11.27. In this example, the appropriate treatment for an asphalt-surfaced road is determined by the pavement condition and the amount of the curb height that is full. Similar decision trees

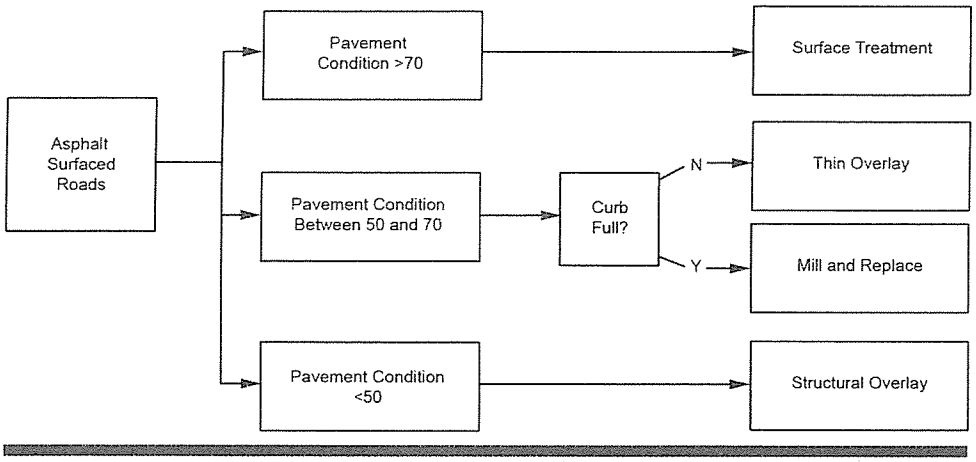


Figure 11.27 Sample Decision Tree

would be developed for other pavement types included in the pavement management database. In reality, decision trees are more complex than the example included in this document, taking into account factors such as traffic, pavement structure, types of distress, geometrics, and other factors in determining appropriate treatment.

Decision Matrices

Another format used for defining treatment rules is a decision matrix. Decision matrices are similar to decision trees in the way they distinguish between treatment options, except that the decisions are presented in a tabular format rather than in a tree structure. The information from Figure 11.27 is illustrated in a matrix format Table 11.1.

Table 11.1
Sample Decision Matrix

Treatment Type	Pavement Condition	Available Curb Height
Surface Treatment	> 70	Any
Thin Overlay	50–70	> 3 in
Mill and Replace	50–70	< 3 in
Structural Overlay	**< 50**	**Any**

Other Tools

Depending on the type of pavement management system used, other tools may be used to determine when various treatments are feasible. Some pavement management systems require the rules to be defined in terms of "if-then" statements that can be easily interpreted by the computer during the analysis. Regardless of the format used, the treatment rules should reflect current agency policies and practices as much as possible so the treatment recommendations make sense.

Benefits and Limitations of Tools

There are advantages and disadvantages to each approach used to determine appropriate treatments in a pavement management system. All approaches are similar in that they define the conditions that must exist before a treatment is considered feasible in the analysis. Some agencies prefer the decision-tree approach because it is generally easier to explain than "if-then" statements. However, they can become quite detailed and difficult to draw when many factors are taken into account.

There are several limitations to the use of these tools that should be considered by any agency establishing a pavement management system. First, in some cases, treatment rules are defined in a manner so that only one given treatment is appropriate for a given set of conditions. While this might sometimes be true, in many cases, multiple treatments may in fact be appropriate, and the pavement management system should determine the cost-effectiveness of each strategy. Therefore, the treatment rules should be set up to allow multiple treatments to be selected when appropriate.

Second, the agency has to be sure that no conditions are excluded in the analysis. This is a common mistake made when defining treatment rules because of gaps in the rules. For instance, one rule might define a treatment for pavement conditions between 50 and 70 and another might define treatments for pavement conditions between 75 and 90. A gap occurs between the pavement conditions 70 to 75 where no treatment is defined.

Finally, because treatment practices can change within an agency, it is important that the treatment rules and cost be reviewed at least annually to ensure that they reflect current conditions. Otherwise, the pavement management system could be making treatment recommendations that do not reflect agency practices or estimating the cost of the treatment erroneously.

Determining Treatment Effectiveness and Timing

The effectiveness of any maintenance or rehabilitation treatment is dependent on its design, construction, and timing. This is especially true with preventive maintenance treatments because there is a narrow window during which they can be applied effectively. A pavement management system can help an agency determine the most cost-effective time to place various treatments using the benefit/cost ratio described earlier.

Developing Guidelines for Treatment Selection

To assist agencies in consistently using treatments effectively, a number of agencies have developed guidelines for treatment selection that assist field personnel in choosing projects and that guide construction inspection procedures. These guidelines define the conditions that might exist for a treatment to be considered a feasible alternative, including distress type, severity, and extent. Photos of appropriate pavement surfaces for each treatment might also be included in the guidelines to help field personnel better understand when a pavement section is considered appropriate.

11.4.3 Types of Maintenance and Rehabilitation Treatments

Three general treatment categories that are considered for a pavement depending on the type and amount of deterioration present: preventive maintenance, rehabilitation, and reconstruction. A general description of each level of repair that might be considered is presented here. Further information on pavement maintenance and rehabilitation treatments can be found in Chapter 12, "Maintenance of Asphalt Pavements" and Chapter 13, "Rehabilitation of Asphalt Pavements" respectively.

Preventive Maintenance

Preventive maintenance treatments are generally applied to pavements in good condition and without much structural deterioration. They include a variety of low-cost treatments aimed at addressing specific types of functional deterioration or slowing the rate of pavement deterioration. Examples of preventive maintenance treatments include crack sealing, surface seals, thin hot mix overlays, and microsurfacing.

Rehabilitation

As pavement deterioration progresses, more substantial treatments may become necessary. Treatments included in this category extend the life of existing pavement structures by either restoring existing structural capacity or by increasing pavement thickness to accommodate existing or projected traffic loading conditions. Pavement rehabilitation may be considered minor rehabilitation or major rehabilitation. Minor rehabilitation treatments include non-structural enhancements, such as thin functional overlays. Major rehabilitation, on the other hand, adds structural enhancements to a pavement section. An example of a major overlay treatment is a thick overlay.

Reconstruction

Reconstruction is the replacement of the entire existing pavement structure, typically including the removal and replacement of all pavement layers including the base. New or recycled materials may be used in a reconstructed pavement section. Reconstruction is the most substantial level of repair considered in a pavement management system. It is recommended on pavement sections that are considered to be in failed condition.

11.4.4 Prioritizing Project Needs

Most agencies are faced with the challenge of having insufficient funds to address all their pavement needs. Pavement management systems assist agencies in this situation by determining the most cost-effective use of available funds. A number of different techniques can be used to determine the optimal use of funds, as discussed in the following sections.

Approaches Used

Three approaches are generally used to prioritize pavement needs: ranking, benefit/cost analysis, and optimization. Each technique is discussed here in more detail in increasing order of sophistication.

Ranking

The simplest method of prioritizing project needs is to rank them according to agency-specified criteria. In most agencies that use a ranking technique, pavement needs are ranked from those in worst condition to those in the best condition. Along with the ranked listing are treatment costs so that projects can be selected from the ranked list until funding is depleted. In some instances, traffic volumes are used to supplement the ranked list so that higher volume facilities are selected for the program before facilities with the same pavement condition but with lower traffic volumes. Since most agencies using a ranking technique are doing so to fix the worst roads first, this approach often results in the least cost-effective strategy for managing a pavement network. This is especially true because alternative strategies (such as preventive maintenance strategies) are not typically incorporated into the ranked list.

Benefit/Cost Analysis

A slightly more sophisticated approach for identifying cost-effective projects and treatments is a multi-year benefit/cost analysis. Under this technique, all feasible treatments over the analysis period are considered in terms of the increased benefit provided by the treatment and the associated cost of the treatment. In most pavement management systems, benefit is defined as the additional performance provided by a treatment, as shown in Figure 11.28. Some agencies multiply the area under the performance curve by a traffic factor so that roads with higher traffic volumes are given more "benefit" than roads with the same pavement condition but with lower traffic volumes. The benefit value is divided by the cost of the treatment to

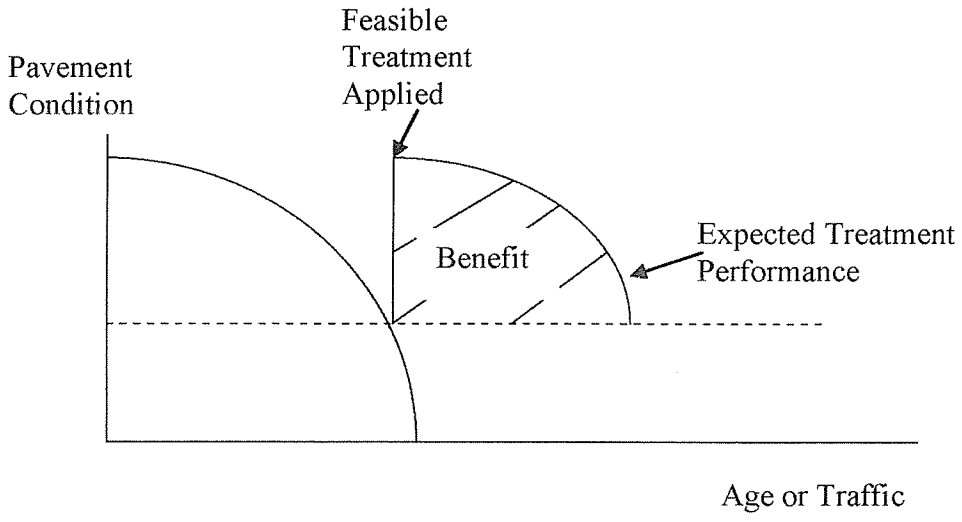


Figure 11.28 Sample Benefit Calculation

determine a benefit/cost ratio. Higher benefit/cost ratios represent a more cost-effective treatment strategy for a given pavement section, treatment type, and treatment timing. Under this type of analysis, the benefit/cost ratios are used to determine the most cost-effective use of the funds available.

Optimization

The most sophisticated approach for determining the optimal program uses mathematical programming techniques as part of a two-step optimization process. First, the objectives for the analysis are defined in mathematical terms, such as maximizing network conditions. Any constraints to the analysis are also defined, such as limits on funding available or minimum allowable conditions. Using mathematical programming techniques such as linear programming or dynamic programming, the optimal solution for the problem set is determined. At this first stage, recommendations are stated in terms of the number of miles to move from one condition category to another.

The second stage of the analysis matches pavement sections to the recommended strategy. For instance, if the optimal strategy indicates that 20 miles of roads in poor condition should be moved into good condition (through a treatment such as reconstruction), then the agency must identify the 20 miles of poor pavement that will be included in the program.

Because of its complexity, a true optimization analysis is used by only a few large agencies.

11.4.5 Maintaining the Pavement Preservation Program

To ensure the success of a pavement preservation program, the preservation philosophy should be integrated into the agency's decisions at all levels. Several keys to the success of an ongoing pavement preservation program are listed here.

Support of Upper Management

The support of upper management is critical to the success of a pavement preservation program, especially in the provision of the funds needed for pavement preservation activities. Upper management support is also important when outside pressure for fixing the worst roads first is applied. Using a pavement management system, upper management can demonstrate the negative long-term impacts of projects initiated through political or other outside pressure and often minimize the inclusion in the program.

Integrating the Philosophy into the Agency

A pavement preservation program often represents a change in the way an agency does business. Rather than strictly focusing on roads in poor condition, the agency adopts a more cost-effective strategy that is sometimes hard for personnel to understand without training. One of the

most effective ways of integrating pavement preservation strategies into decisions is to demonstrate the cost-effectiveness of the program and the anticipated conditions that will result. Personnel at all levels of the agency must understand and buy into the philosophy for it to be most effective.

Demonstrating Benefits

The benefits associated with a pavement preservation program can be demonstrated in a number of ways. Most agencies focus on the expected improvement in pavement conditions and safety for the traveling public. However, it may be more effective to convey the cost-effectiveness of a pavement preservation program to politicians and upper management because they often have business backgrounds rather than engineering backgrounds. For example, an agency may demonstrate the cost of achieving a targeted condition level under both a worst-first strategy and a pavement preservation strategy. In most instances, the cost of achieving the same condition under a worst-first strategy is four to six times more expensive than it would be under a pavement preservation strategy.

Importance of Training

The importance of communicating the pavement preservation philosophy cannot be overstated. To be fully integrated within the agency, it is important that personnel understand the underlying philosophy and their role in supporting the agency's pavement preservation efforts. A number of training programs on pavement preservation techniques are available through the Asphalt Institute, the American Society of Civil Engineers, and the FHWA's National Highway Institute.

Resources

1. American Association of State Highway and Transportation Officials. 1993. *Guide for Design of Pavement Structures.* American Association of State Highway and Transportation Officials. Washington, DC.
2. Asphalt Institute. *Asphalt in Pavement Maintenance,* MS-16. The Asphalt Institute. Lexington, KY.
3. Asphalt Institute. 2000. *Asphalt Overlays for Highway and Street Rehabilitation,* MS-17. The Asphalt Institute. Lexington, KY.
4. Asphalt Institute. 2001. *A Pavement Rating System for Low-Volume Asphalt Roads,* Information Series No. 169. The Asphalt Institute. Lexington, KY.
5. Federal Highway Administration. 1999a. *Conditions and Performance Report.* Federal Highway Administration. Washington, DC.
6. Federal Highway Administration. 1999b. *"Pavement Preservation: A Road Map for the Future."* FHWA-SA-99–015, Forum held October 26–28, 1998. Kansas City, MO.
7. Federal Highway Administration. 2003. *Distress Identification Manual for the Long-Term Pavement Performance Program.* Federal Highway Administration. Washington, DC.
8. Dave Geiger, memorandum on pavement preservation definitions, Sept. 12, 2005. Federal Highway Administration, Washington, DC.